Activity 1.2: Logic & Reasoning

A

Name: _____ Partner's name: _____ Date: _____

Directions: Answer each question. For true/false questions, give a counterexample if the statement is false. Problems 5 and 6 ask you to name a law of logic.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

Find the next term: 2, 6, 18, 54, ...

Answer: _____ Match? ☐ Yes ☐ No

2

Find the next term: 3, 7, 11, 15, ...

Answer: _____ Match? ☐ Yes ☐ No

3

Statement: If $x^2 > 9$, then $x > 3$. Which value is a counterexample: $x = 4$, $x = -5$, $x = 3.5$, or $x = 10$?

Answer: _____ Match? ☐ Yes ☐ No

4

Is the *converse* of "If a figure is a square, then it is a rectangle" true or false? If false, give a counterexample.

Answer: _____ Match? ☐ Yes ☐ No

5

"If a number is divisible by 6, then it is divisible by 3. 18 is divisible by 6. Therefore, 18 is divisible by 3." Which law of logic is used: Detachment or Syllogism?

Answer: _____ Match? ☐ Yes ☐ No

6

"If it rains, the field is wet. If the field is wet, practice is cancelled. Therefore, if it rains, practice is cancelled." Which law of logic is used: Detachment or Syllogism?

Answer: _____ Match? ☐ Yes ☐ No

7

Solve $3(x-2) = 12$ by first rewriting the left side as $3x - 6$. Name the property used for that step and find x.

Answer: _____ Match? ☐ Yes ☐ No

8

$\angle 1$ and $\angle 2$ are supplementary and $m\angle 1 = 70°$. Find $m\angle 2$.

Answer: _____ Match? ☐ Yes ☐ No

Activity 1.2: Logic & Reasoning

B

Name: _____ Partner's name: _____ Date: _____

Directions: Answer each question. For true/false questions, give a counterexample if the statement is false. Problems 5 and 6 ask you to name a law of logic.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

The sequence 6, 18, 54, ... continues the same pattern. Find the 4th term.

Answer: _____ Match? ☐ Yes ☐ No

2

Find the next term: 4, 7, 10, 13, 16, ...

Answer: _____ Match? ☐ Yes ☐ No

3

Statement: If $x^2 > 16$, then $x > 4$. Which value is a counterexample: $x = 5$, $x = -5$, $x = 4.5$, or $x = 8$?

Answer: _____ Match? ☐ Yes ☐ No

4

Is the *inverse* of "If a figure is a square, then it is a rectangle" true or false? If false, give a counterexample.

Answer: _____ Match? ☐ Yes ☐ No

5

"If an animal is a dog, then it is a mammal. Rex is a dog. Therefore, Rex is a mammal." Which law of logic is used: Detachment or Syllogism?

Answer: _____ Match? ☐ Yes ☐ No

6

"If you study, you will learn. If you learn, you will succeed. Therefore, if you study, you will succeed." Which law of logic is used: Detachment or Syllogism?

Answer: _____ Match? ☐ Yes ☐ No

7

Solve $2(x+1) = 14$ by first rewriting the left side as $2x + 2$. Name the property used for that step and find x.

Answer: _____ Match? ☐ Yes ☐ No

8

$\angle 3$ and $\angle 4$ form a linear pair and $m\angle 3 = 70°$. Find $m\angle 4$.

Answer: _____ Match? ☐ Yes ☐ No